In [1]:
```python
import pandas as pd
import numpy as np
import joblib

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

from xgboost import XGBClassifier

RANDOM_STATE = 54895
```

In [7]:
```python
# Load data
df = pd.read_csv("../data/Crop_Yield_Fertilizer.csv")
```

In [8]:
```python
df.head()
```

Out[8]:

	N	P	K	temperature	humidity	ph	rainfall	label	yie
0	90.0	42.0	43.0	20.879744	82.002744	6.502985	202.935536	rice	71.1994
1	85.0	58.0	41.0	21.770462	80.319644	7.038096	226.655537	rice	81.6201
2	60.0	55.0	44.0	23.004459	82.320763	7.840207	263.964248	rice	80.4731
3	74.0	35.0	40.0	26.491096	80.158363	6.980401	242.864034	rice	75.1781
4	78.0	42.0	42.0	20.130175	81.604873	7.628473	262.717340	rice	75.4855

In [9]:
```python
FEATURES = ['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall']
TARGET = ['label']

X = df[FEATURES].copy()
y = df[TARGET].copy()
```

In [10]:
```python
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)
```

```
C:\Users\preda\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5
n2kfra8p0\LocalCache\local-packages\Python311\site-packages\sklearn\preprocessi
ng\_label.py:110: DataConversionWarning: A column-vector y was passed when a 1d
array was expected. Please change the shape of y to (n_samples, ), for example
using ravel().
  y = column_or_1d(y, warn=True)
```

In [11]:
```python
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), FEATURES)
    ]
)
```

In [13]:
```python
model = XGBClassifier(
```

```python
    n_estimators=600,
    max_depth=5,
    learning_rate=0.08,
    subsample=0.9,
    colsample_bytree=0.9,
    objective='multi:softprob',
    num_class=len(label_encoder.classes_),
    eval_metric='mlogloss',
    random_state=RANDOM_STATE
    # n_jobs=-1
)
```

In [14]:

```python
pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('model', model)
])
```

In [15]:

```python
pipeline.fit(X, y_encoded)
```

Out[15]:

▸ **Pipeline**

▸ **preprocessor: ColumnTransformer**

▸ **num**

▸ StandardScaler

▸ XGBClassifier

In [17]:

```python
joblib.dump(pipeline, "../models/crop_model.pkl")
joblib.dump(label_encoder, "../models/crop_label_encoder.pkl")

print("Crop model trained and saved")
```

```
Crop model trained and saved
```